

ABSTRACT

Bitcoin's Proof-of-Work consumes ~150 TWh/year while delivering only ~7 TPS; Ethereum Layer-1 still delivers only 15–30 TPS. Proof-of-Space replaces wasted hashing with disk-backed proofs, but production deployments remain below ~30 TPS because consensus rounds spend too much time coordinating. We present **MEMO**, a Proof-of-Space blockchain implemented from scratch in C with a five-component ZeroMQ architecture and a *deadline-aware* block-creation pipeline. The winning validator checks the remaining round budget before each verify batch and commits a partial block rather than overshooting the metronome tick. On a single 16-core node MEMO reaches **21,758 end-to-end TPS**; across five torus nodes it sustains **17,797 TPS (15,909 end-to-end TPS)** with **100% confirmation**.

MOTIVATION

- **Throughput is still the bottleneck:** Bitcoin is ~7 TPS; Ethereum L1 is 15–30 TPS, far below payment demand.
- **Proof-of-Space is energy-efficient but slow:** disk-backed proofs reduce waste, yet production PoSpace remains below ~30 TPS.
- **Centralized rails set the target:** Visa-scale systems sustain ~65,000 TPS, leaving trustless systems orders of magnitude behind.
- **MEMO attacks systems overhead:** strict deadlines, winner-only fetch, and batched verification turn idle round time into committed TXs.

TESTBED

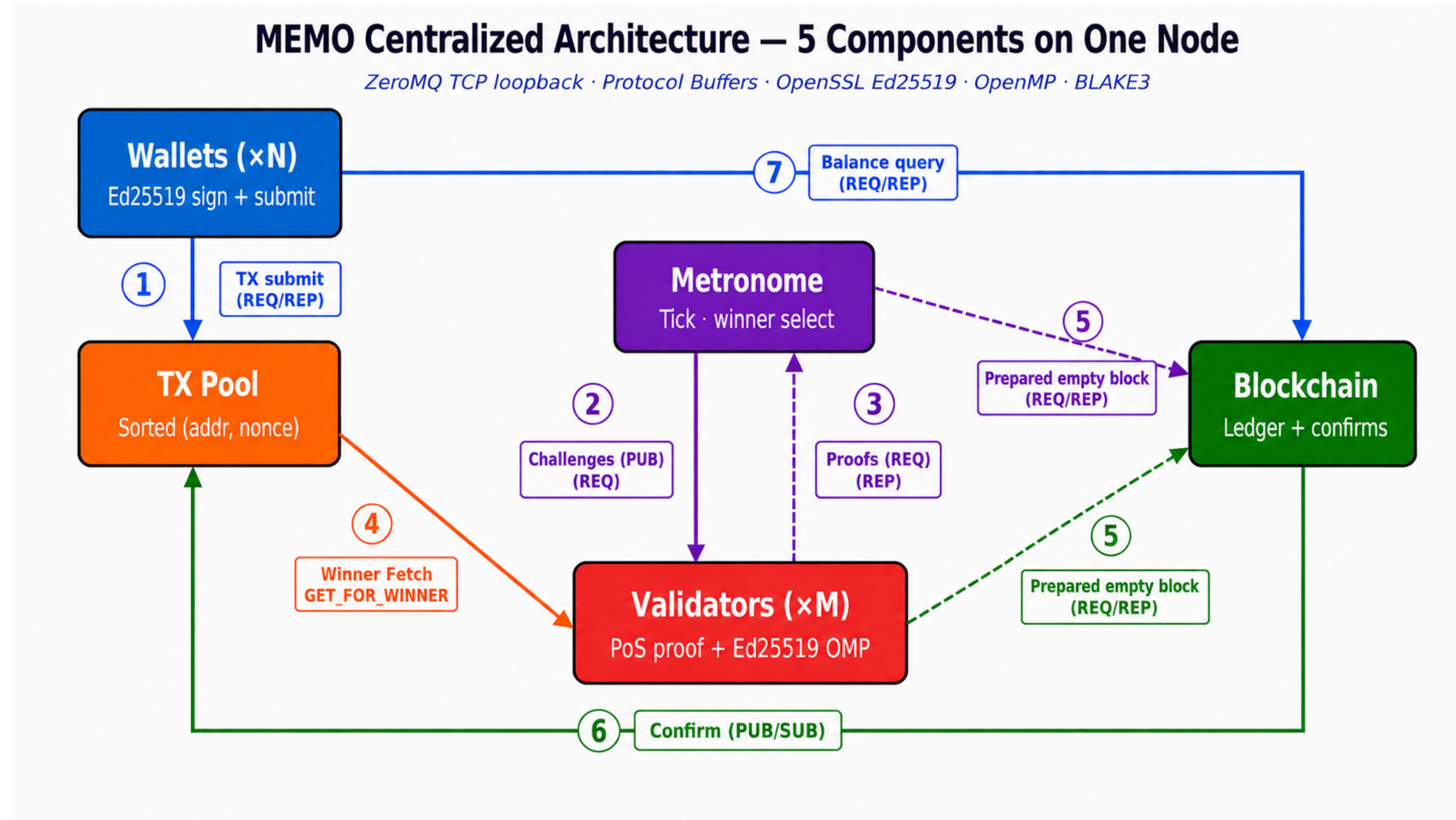
	Nodes	Sockets	Processor	Memory	Network	Threads
Centralized	1	2	Intel SP 4108 16C @1.8GHz	64GB	100 GbE	up to 32
Distributed	5	2	Intel SP 4108 16C @1.8GHz	64GB	100 GbE	up to 32

- Implemented from scratch in **11,600 LOC of C**.
- **Workload:** 1 M Ed25519-signed TXs/run, 5–100 pre-funded sender wallets.
- Sweeps cover **block size, OpenMP threads, proof window, verify batch size, number of wallets, number of validators**.
- Metrics report confirmed on-chain throughput, not wallet-side submission rate.

KEY DESIGN PRINCIPLES

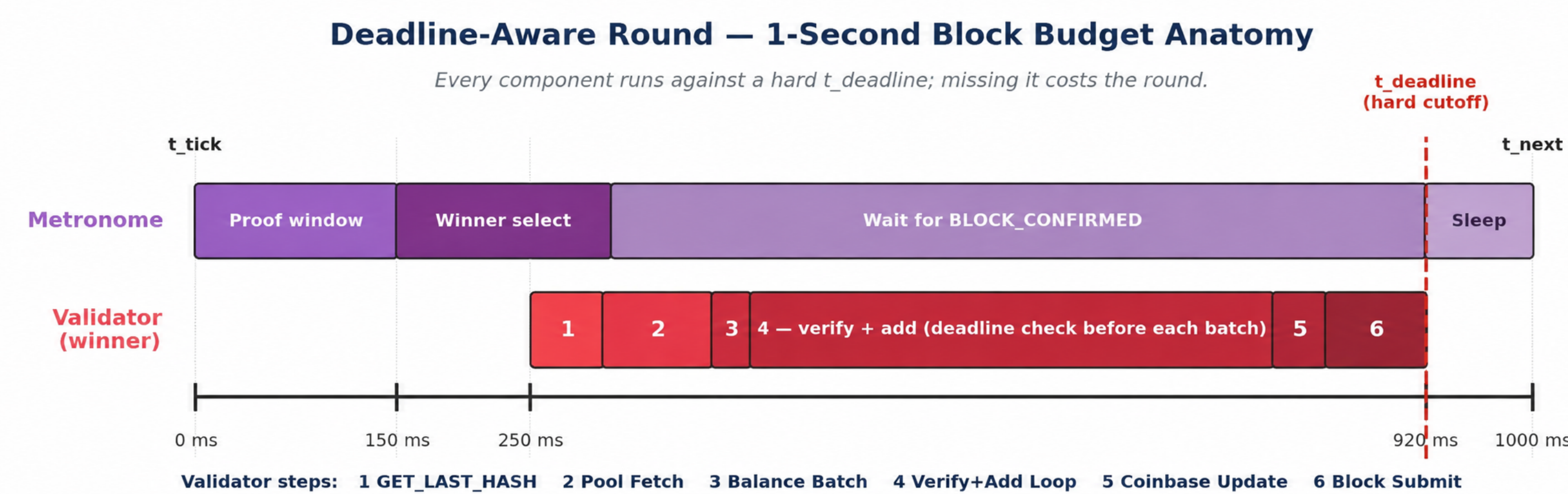
- **Winner-only fetch:** only the selected validator pulls sorted TXs; non-winners avoid duplicate work.
- **Budgeted fetch cap:** the validator fetches only what the remaining deadline can plausibly verify.
- **Batched balance + verify:** balances are queried once; Ed25519 verification runs in OpenMP batches.
- **Partial-block fallback:** if the deadline nears, MEMO commits completed TXs instead of overshooting.
- **Validator count decoupling:** extra validators compete during proof collection, not during block construction.

ARCHITECTURE



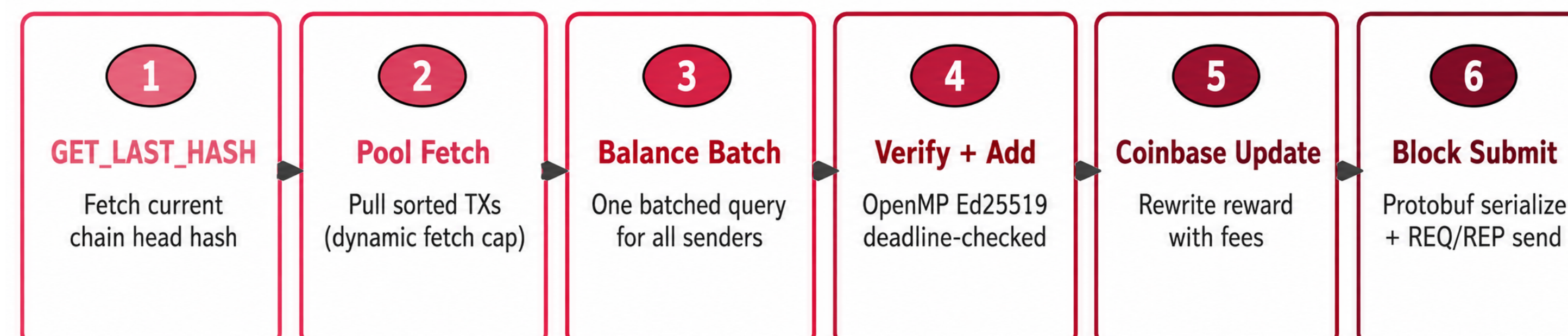
- Centralized and distributed deployments use the same protocols; only process placement changes.
- Wallets submit signed TXs, the metronome selects a winner, and the winning validator commits the prepared block.

DEADLINE-AWARE DESIGN



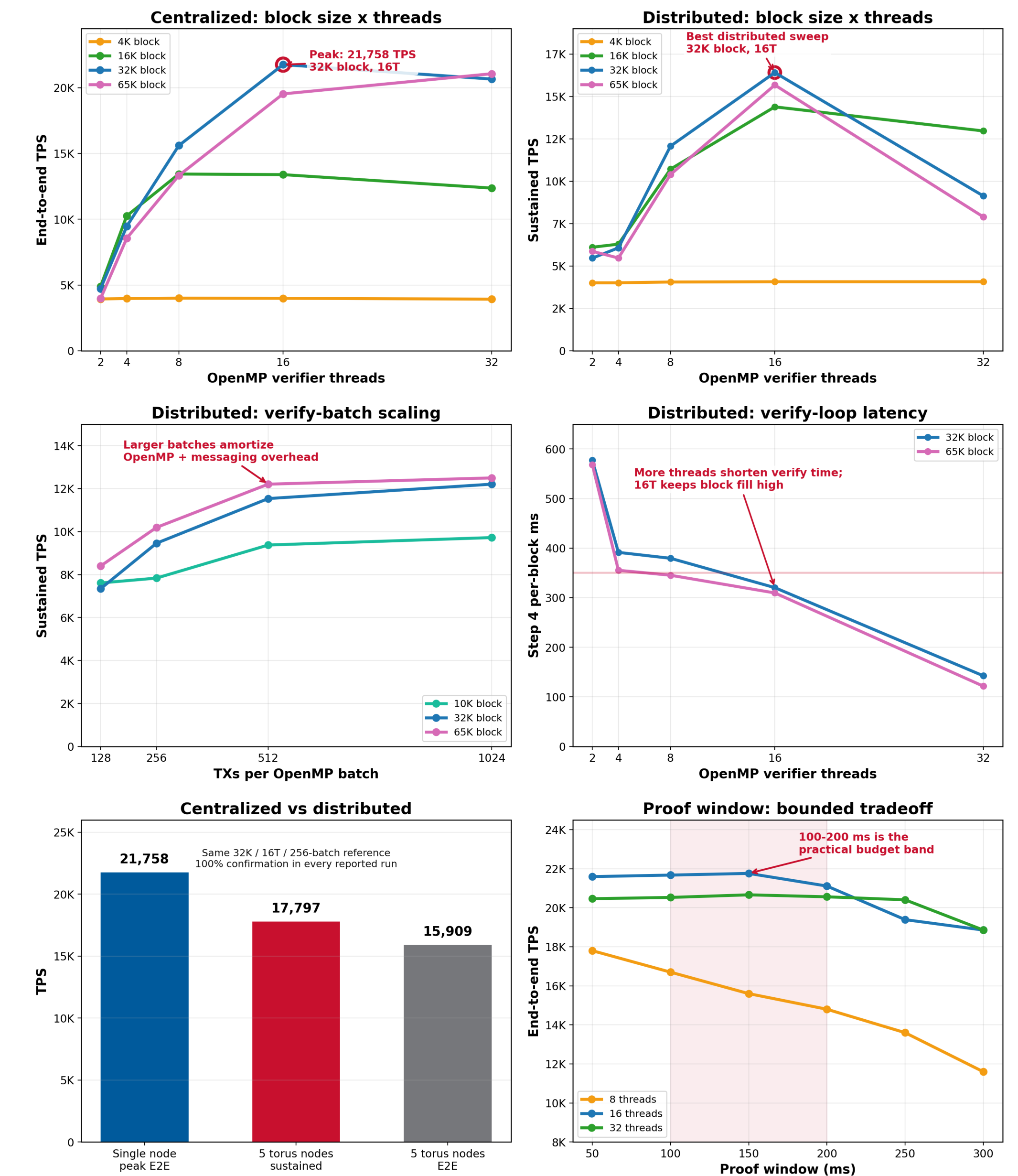
- A hard t_{deadline} bounds every round; completed TXs are submitted before cutoff.
- The proof window trades winner quality against verification time; 100–200 ms is the measured budget.

Validator's 6-Step Pipeline (executed by the round's winner)



- Step 4 dominates runtime, so MEMO batches verification and checks the deadline before each batch.

RESULTS



- **Centralized peak:** 32K block, 16 verifier threads, 256 verify batch $\Rightarrow$ **21,758 TPS**.
- **Distributed result:** same reference config sustains **17,797 TPS**; end-to-end run rate is **15,909 TPS**.
- **Distributed trend:** 16T gives the best block fill; 32T verifies faster but loses throughput to overhead and underfilled rounds.

FUTURE WORK

- **WAN-scale consensus:** 100–1,000 nodes, sharded pools, per-node validators, multi-metronome scheduling.
- **Persistent storage:** evaluate XStore (log-structured), RocksDB (LSM tree), LMDB (B+ tree) for crash-safe ledger persistence.
- **Post-quantum signatures:** evaluate Dilithium, SPHINCS+, and Falcon-512 signatures for forward-looking security.

CONCLUSION

- **21,758 TPS centralized; 17,797 sustained TPS** across five torus nodes.
- **100% confirmation** across the reported sweeps; no empty rounds from deadline overrun.
- The main lever is **deadline-aware systems engineering**: sorted fetch, batched verify, partial-block commit.
- **Pure C + ZeroMQ + OpenMP** — no exotic crypto, no GPU.